

be the agent

artificial intelligence · prof. dr. nicolas meseth

- 1 the game
- 2 play
- 3 what you just played

now, you
are the agent,
and the computer
is your group.



part 1

the game

seven roles

model

sees only its stack of cards

seven roles

search

web

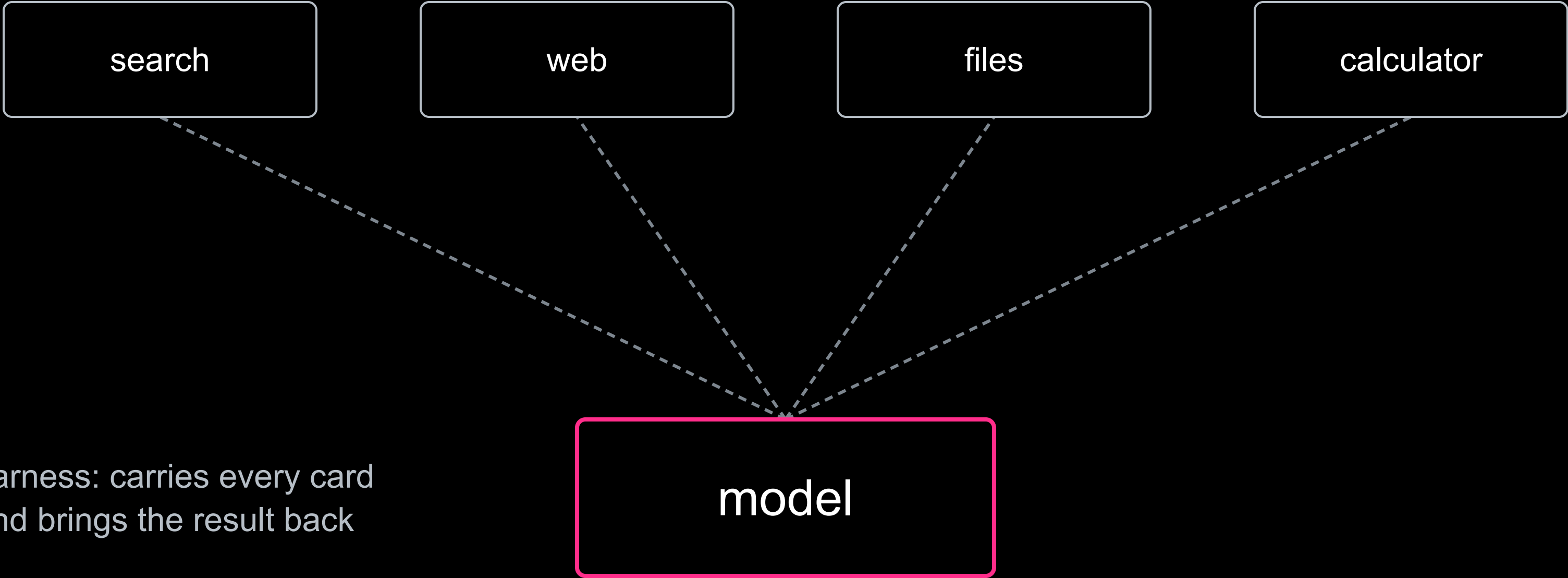
files

calculator

model

sees only its stack of cards

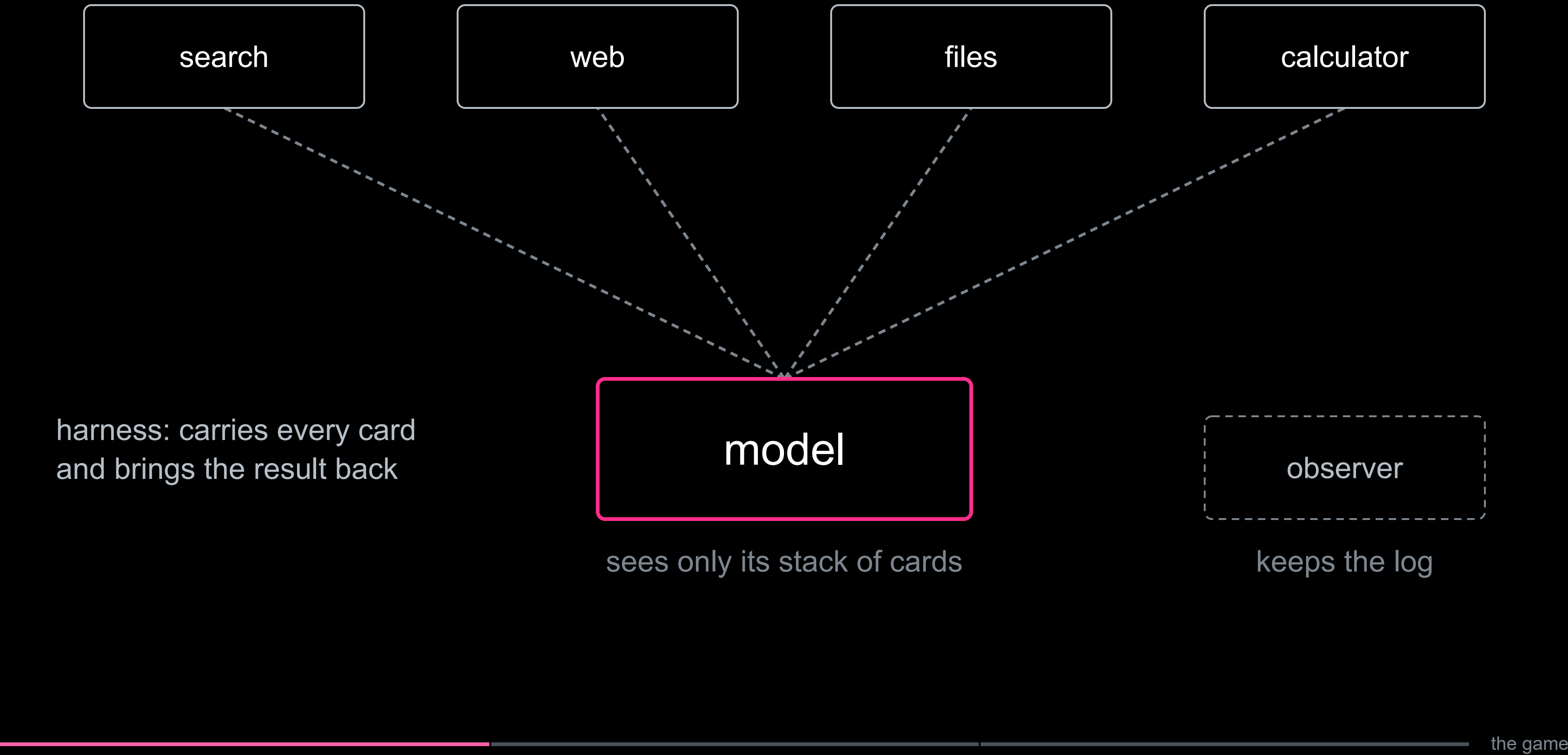
seven roles



harness: carries every card
and brings the result back

sees only its stack of cards

seven roles



four kinds of cards

THINK :

which files
are there?

four kinds of cards

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TOOL :

```
list_files()
```

four kinds of cards

THINK :

which files
are there?

TOOL :

`list_files()`

RESULT :

fields,
soil_tests,
stock, weather

four kinds of cards



the rules

the model sees only its stack, and nobody talks to it.

the game

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the rules

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the harness carries every card to the right tool and brings the result back.

a tool does exactly what the card says, nothing more.

one card at a time, until the model writes ANSWER :.

the system card

<code>search("words")</code>	finds web pages and returns their titles
<code>open("title")</code>	returns the full text of one web page
<code>list_files()</code>	returns the names of all farm files
<code>read_file("name")</code>	returns the text of one farm file
<code>calculator("expression")</code>	calculates, e.g. <code>calculator("4.5 * 100")</code>
<code>write_file("name", "text")</code>	saves a text as a new farm file

this is everything the model knows about its tools.

part 2

play

task 1 · about 5 minutes

how large is the field long acre, and what grows on it this year?

task 2 · swap roles · about 12 minutes

how much nitrogen may we still apply to long acre this spring? give the total in kg N for the whole field.

which nitrogen requirement did your model use, and where did it come from?

task 3 · if there is time

plan the first nitrogen application on long acre: how much of which product do we need, and can we spread it on 10 february?

part 3

what you just played

every card
on this table
is one step
of the agent.

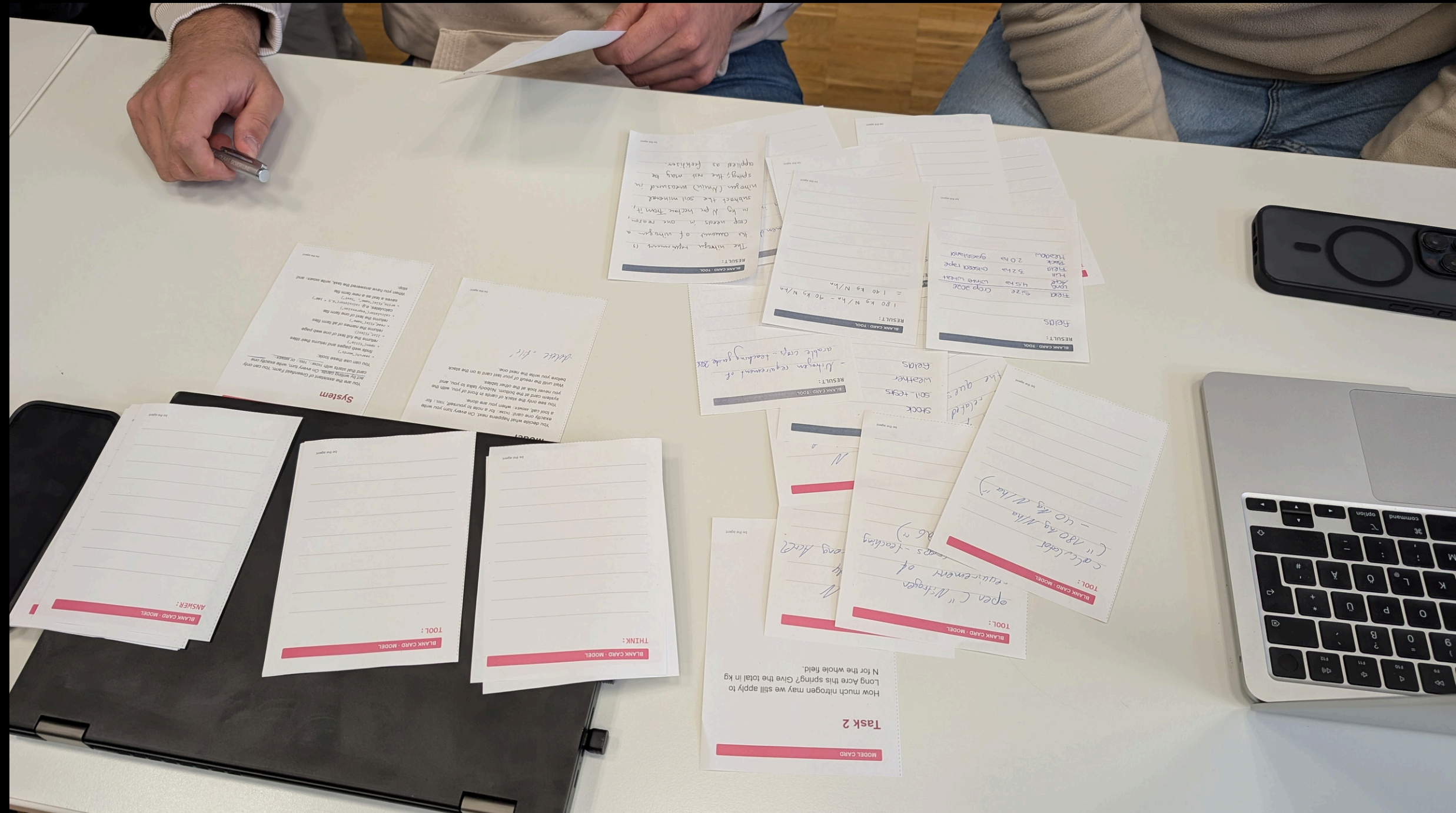


photo: a group playing task 2

what you just played

from the game to a real agent

the stack of cards → **context window**: everything the model knows

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RESULT : → **observation**, added to the context

from the game to a real agent

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the system card	→	system prompt : who it is, which tools it has
THINK :	→	reasoning , the plan
TOOL :	→	tool call , written as text
the harness	→	harness : the program that runs the tools
RESULT :	→	observation , added to the context
ANSWER :	→	stop : the goal is reached

the same, in a real agent

TOOL	list the files in the folder <code>Get-ChildItem -Recurse</code>
RESULT	one file: a previous group's report ... <code>Applications of Artificial Intelligence.md</code>
TOOL	read that file <code>Read("... .md")</code>
RESULT	“a brief history of artificial intelligence ... 1950: alan turing ...”
THINK	“before the plan, two things are missing: there's no pinboard photo, and there's no file called milestones.md.”
TOOL	start eight sub-agents <code>Agent("G1 foundations 1943–1969") ...</code>

an AI agent is a **large language model (LLM)** that can only write text; everything it does in the world happens through tools that the **harness** runs for it.

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who did the thinking, who did the acting, and which of it could the model actually do?